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ABSTRACT

abstract

Keywords: word1 — word2 — word3 — word4 — word5

INTRODUCTION

The PageRank algorithm, first introduced by Larry Page and Sergey Brin, was one of the first algorithms to rank webpages based on their link structure rather than just content Brin & Page (1998). The algorithm models the web as a directed graph and simulates a "random surfer" who follows hyperlinks with some probability or jumps to a random page to avoid getting stuck Page et al. (1999).

This setup defines a Markov process where each page's importance is proportaional to the importance of pages linking to it. The final PageRank vector corresponds to the steady-state distribution of this process and can be found either by iteration or by solving for the dominant eigenvector of the hyperlink matrix $\boldsymbol{H}$ Langville & Meyer (2009).

In this project, we first implemented PageRank using the difference equation $x_{k+1} = \boldsymbol{H} x_k$, which resulted in the scores converging once the solution reached the maximum tolerance allowed. We ten confirmed our result by solving for the eigenvector associated with the largest eigenvalue of the matrix $\boldsymbol{H}$. Our goal was to understand how the algorithm behaves under different network structures and how quickly it converges Langville & Meyer (2004).

DATA AND OBSERVATIONS

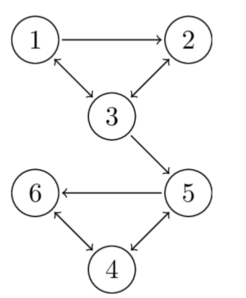

Figure 1. example graph 1.

To start, we defined a 6-node directed graph where each page links to one or more pages, as shown in Figure 1. From this, we constructed the hyperlink matrix $\boldsymbol{H}$, where each non-zero entry $\boldsymbol{H}_{ij}$ is equal to $\frac{1}{|P_{ij}}$, the reciprocal of the number of outbound links from page P_j. This matrix represents the probability that a random surfer on page j clicks through to page i:

$$
\boldsymbol{H} = \begin{pmatrix} 0 & 0 & 1/3 & 0 & 0 & 0 \\ 1/2 & 0 & 1/3 & 0 & 0 & 0 \\ 1/2 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/2 & 1 \\ 0 & 0 & 1/3 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 1/2 & 1/2 & 0 \end{pmatrix} \tag{1}
$$

We initialized the PageRank vector $\boldsymbol{x}_0$ with equal values and assumed that each page is equally likely to be visited at first:

$$
\boldsymbol{x}_0 = \begin{pmatrix} 1/6 \\ 1/6 \\ 1/6 \\ 1/6 \\ 1/6 \\ 1/6 \end{pmatrix} \tag{2}
$$

We then applied the iterative update rule $\boldsymbol{x}_{n+1} = \boldsymbol{H}\boldsymbol{x}_n$ until the rank vector converged below a given tolerance to simulate a random surfer moving moving through the web graph via the links. We visualized the convergence of the PageRank vector by plotting each page's rank score for each iteration, which allowed us to track how the scores changed until they reached a steady state. The resulting PageRank scores are:

$$
\boldsymbol{x}_{\text{converged}} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 4/9 \\ 2/9 \\ 1/3 \end{pmatrix} \tag{3}
$$

NOTE: REWRITE THIS PART IT SOUNDS WEIRD Pages 4–6 dominate the steady-state distribution, whereas Pages 1–3 decay to zero in the same limit. Page 4 having the highest PageRank score of 4/9 means it is the most frequently visited node and most dominant page in the network. This behavior matches the convergence pattern shown in Figure 2.

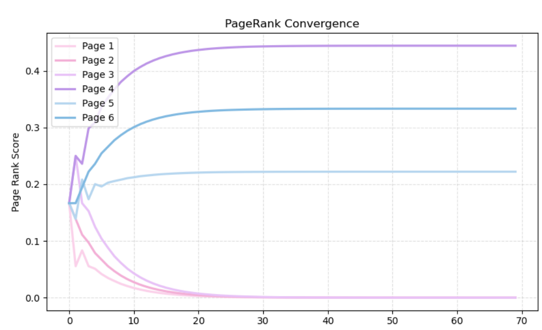

Figure 2. PageRank convergence for graph 1.

TALK ABOUT SECOND GRAPH EX HERE

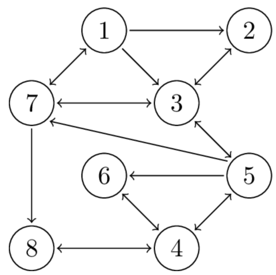

Figure 3. example graph 2.

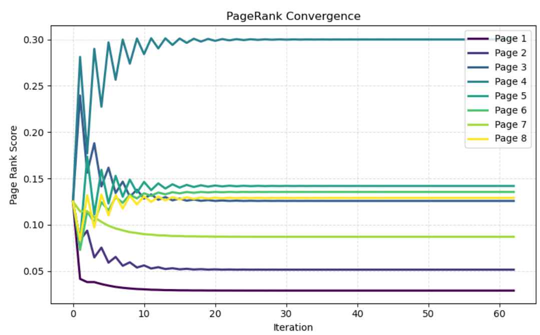

Figure 4. PageRank Convergence for graph 2.

RESULTS

results

SUMMARY AND CONCLUSION

summary/concl

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